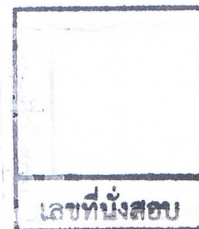


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2013

Subject: 392151 Chemistry I

Section: 15 - 16

Date: 1 August 2013

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Textbook and dictionaries are **NOT** allowed.
 4. Calculators are allowed.
 5. Electronic communication devices are **NOT** allowed in the examination room.
 6. The examination has 8 pages (including this page), 2 sections and a total score of 56 points.
 7. Write all your solution and answer on this examination sheet.
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Part 1. Multiple choice (1.5 points each)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				

Question no.	a.	b.	c.	d.
8				
9				
10				
11				
12				
13				
14				

- Which of the following consists of one element?
 - Carbon in pencil
 - Carbon monoxide
 - Syrup
 - Pure water
- Which two words apply to the sodium chloride (salt)?
 - liquid, mixture
 - solid, mixture
 - liquid, compound
 - solid, compound
- Which of the following is a chemical change?
 - Ice starts to melt
 - A car starts to rust
 - Warming food in microwave
 - Separating sand and gravel
- Which method of separation is best to use for separating water containing small quantities of cooking oil?
 - Filtration
 - Centrifugation
 - Distillation
 - Evaporation
- Neutron was discovered by _____.
 - Chadwick
 - Thomson
 - Rutherford
 - Bohr
- The K, L and M shells of an atom are full. Its atomic number is _____.
 - 28
 - 20
 - 18
 - 10

7. The mass of the atom is determined by _____.
a. neutron
b. neutron and proton
c. electron
d. electron and neutron
8. Which atomic model was the first one to include a nucleus?
a. Dalton's billiard ball model
b. Rutherford's electron cloud model
c. Thomson's plum pudding model
d. Bohr's planetary model
9. The cathode ray tube was instrumental in the discovery of:
a. the electron
b. the neutron
c. the proton
d. the atom, which is mostly empty space
10. Which particles have approximately the same size and mass as each other?
a. neutrons and electrons
b. electrons and protons
c. protons and neutrons
d. none - they are all very different in size and mass
11. What is the key difference between the models of the atom proposed by Bohr and Rutherford?
a. Bohr's model includes a nucleus, but Rutherford's does not.
b. Bohr's model has the atom made up of mostly empty space, but Rutherford's does not.
c. Bohr's model includes neutrons, but Rutherford's does not.
d. Bohr's model has the electrons moving in fixed energy levels, while in Rutherford's they do not.
12. Consider the isotope ${}_{92}^{234}\text{U}$. Which of the following statements is true?
a. The element is an isotope of ${}_{94}^{234}\text{Pu}$.
b. The element contains 234 neutrons.
c. The element has the same electron configuration as ${}_{92}^{233}\text{U}$.
d. The element has an atomic mass number of 92.
13. Changing the number of neutrons of an atom changes its:
a. isotope
b. element
c. ion
d. charge

14. The electron configuration for helium is $1s^2$. The "1" stands for....
- the orbital
 - how many electrons helium has
 - the energy level the electrons are in
 - the shape of the area that the electron's in 90% of the time

Part 2 Answer (35 points)

1. Classify each of the following mixture as homogeneous or heterogeneous. (2 points)

1.1 vegetable soup _____ 1.2 salt water _____
 1.3 tea with ice _____ 1.4 pizza _____

2. Classify each of the as an element, a compound or a mixture. (2 points)

2.1 carbon in pencil _____ 2.2 ice _____
 2.3 air _____ 2.4 sandwich _____

3. Classify each of the following changes in matter as physical or chemical. (2 points)

3.1 Grinding chalk into powder _____
 3.2 Burning carbon _____
 3.3 Hammering gold into foil _____
 3.4 Digesting food _____

4. Give a symbol of element described by the following: (2 points)

Example Group 8A, Period 1 : He

4.1 A halogen in period 4 _____ 4.2 An alkali metal in period 2 _____
 4.3 Group 2A, Period 3 _____ 4.4 The first transition element in period 4 _____

5. For each of the following reactions, say whether it is an example of a synthesis or decomposition reaction: (2 points)



6. Calculate the frequency and energy of a photon of light of wavelength $4.340 \times 10^{-7} \text{ m}$, corresponding to a line in the visible spectrum of hydrogen. (4 points)

(speed of light; $c = 3.0 \times 10^8 \text{ m/s}$ and Plank's constant; $h = 6.63 \times 10^{-34} \text{ J.s}$)

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7. Complete the following table. (Hint: using detail in the periodic table) (6 points)

Nuclear Symbol	Atomic number (Z)	Atomic mass number (A)	Number of protons	Number of electrons	Number of neutrons
${}^{56}_{26}\text{Fe}$					
			8	10	
			24	24	

8. Write the detailed electron configurations and draw the electron diagram in orbitals of Sc and S. (6 points)

Element	Subshell and electron diagram in orbitals	Shell
${}_3\text{Li}$ (Example)	$1s^2 \quad 2s^1$ ↑↓ ↑	2,1
${}_{21}\text{Sc}$		
${}_{16}\text{S}$		

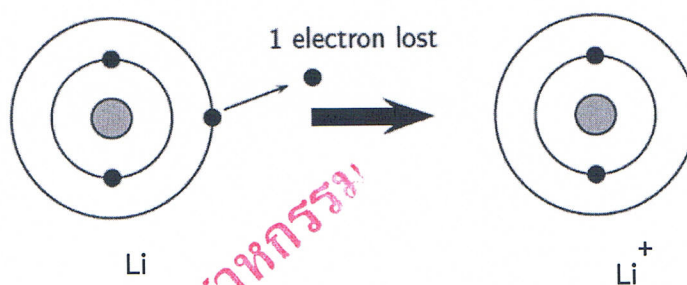
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9. Write the detailed electron configurations and identify its location (period and group) in the periodic table for each of the element in table below (5 points)

Element	Subshell	Shell	Period number	Group number
${}_3\text{X}$ (3electrons)	$1s^2 2s^1$	2,1		
${}_9\text{X}$ (9electrons)				
${}_{31}\text{X}$ (31electrons)				

10. Look at the following examples.

Notice the number of valence electrons in the neutral atom, the number of electrons that are lost or gained, and the final charge of the ion that is formed.



Using the lithium diagram as a guide to draw Fluorine and its ion. (4 points)

Monrudee Phongaksorn